From Words to Worlds

Re-captioning Study: Is the Problem the Model, the Captions, or the 77-Token Limit?

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1. Summary

Stable Diffusion 1.5 often leaves out objects and details that a caption names. This study asks why, using 100 photos from the project's frozen COCO test split. Each photo was captioned four ways: the human COCO caption, a short and a detailed caption from Florence-2-large, and the detailed caption **rewritten to fit** in the 77 tokens Stable Diffusion's text encoder reads. Every caption was turned into an image with the same seed, and every image was compared with the real photo. Four images the safety checker blacked out are left out of all averages. This is a pilot: 100 photos and one seed.

Result (person excluded where marked)	Human · Short · Detailed (cut) · Fitted	What it shows
Photo objects named in the part SD reads (no person)	54.7% · 62.1% · 71.5% · 74.7%	Model captions tell the model more about the scene
Detailed captions over 77 tokens	94% cut off (25 tokens on average) · fitted 0%	Fitting makes the whole caption readable
Similarity of the image to the real photo (CLIP-I)	0.704 · 0.709 · 0.716 · 0.718	A small rise, never detectably above zero
Object recall in generated images (no person)	67.7% · 67.7% · 62.6% · 66.0%	No caption makes the model draw more of the photo's objects
Named-object recall: named, read, drawn (no person)	88.5% · 88.0% · 76.3% · 72.0%	Each object a long caption names is less likely to be drawn
Fitted vs cut-off detailed (key test)	No detectable difference on any of 6 measures	No sign the cut-off was holding detailed captions back
Detector recall on the real photos	91.5%	The gap to 63–68% is what Stable Diffusion leaves out
Named, read, but still missing	79 objects in 73 images (74 trustworthy)	Exported for rescue experiments

Answer, for this sample: longer captions gave no detectable gain. Describing more of the scene, and making sure Stable Diffusion reads all of it, did not detectably bring the images closer to the real photo or make the model draw more of its objects, which points to the model rather than the captions as the main limit. Gains up to the top of each 95% range cannot be ruled out until the study is repeated with 300 photos and two seeds.

2. Question and design

Explanation	If true, the fitted caption should…
A. The cut-off: the useful detail was in the part Stable Diffusion never read	do detectably better than the cut-off detailed caption
B. The model: Stable Diffusion cannot use more detail even when it reads it	do no better than the cut-off detailed caption or the human caption

Step	What was done	Why
Photos	First 100 pairs of the frozen 2,000-pair test split (rebuilt with the Milestone 1 rules and checked against the saved split file), each as its 512 × 512 center crop	The same photos and crop the generated images are compared with

Step	What was done	Why
Captions	Human (COCO); Florence-2-large short (<CAPTION>) and detailed (<MORE_DETAILED_CAPTION>), beam search, no sampling; detailed fitted to 77 tokens (Section 3)	Short isolates the captioner's wording; detailed adds content; fitted removes the cut-off
Generation	Stable Diffusion 1.5, PNDM, 30 steps, guidance 7.5, **the same seed for all four captions of a photo**	The only difference between the four images is the text
Judging	Against **the real photo**: CLIP-I, object recall and named-object recall (Faster R-CNN; person excluded and reported separately); plus CLIPScore with the image's own prompt and with the human caption	Judging against the scene treats every caption equally
Clean-up	Images the safety checker blacked out (all-black) are detected and left out of every average: 4 of 400, one per caption source (three of them the same photo)	A black image would count as a failure on every measure
Statistics	Paired bootstrap 95% ranges over photos (5,000 resamples); a pair is used only if both images have a score	A difference is called detectable only if its range excludes zero

98 of the 100 photos show at least one labelled object after the crop (median 2 object types). The study ran fully from scratch on a Tesla T4 (about 45 minutes); two later runs reused the saved captions and images and reproduced every number.

3. Fitting detailed captions to 77 tokens

CLIP reads 77 tokens, two of which are start and end markers, leaving 75 for the caption. Each detailed caption was rewritten using only its own text, never the photo's labels: 1. split into sentences; 2. remove filler such as "The image shows"; 3. always keep the first sentence (the main subject); 4. add the other sentences greedily, each time the one naming the most COCO objects not yet mentioned, while it still fits; 5. put the kept sentences back in their original order. On average 4.1 of 5.7 sentences were kept, at 69 tokens; every fitted caption fits. The dropped sentences mostly describe weather, lighting, mood or distant background.

Detailed (cut at 77 by the pipeline)	Fitted (69 tokens)
The image shows a baseball game in progress. The batter is at home plate and is in the process of swinging his bat. He is wearing a black jersey with the number 21 on it and a black helmet. The catcher is crouched behind home plate, ready to catch the ball. In the background, there are spectators watching the game from the dugout. The field is covered in green grass and there are trees and …	A baseball game in progress. The batter is at home plate and is in the process of swinging his bat. He is wearing a black jersey with the number 21 on it and a black helmet. The catcher is crouched behind home plate, ready to catch the ball. In the background, there are spectators watching the game from the dugout.

4. How the captions differ

Caption source	Words	CLIP tokens	Over 77	Tokens cut	Photo objects named (all)	No person: whole	No person: part SD reads	Named, not labelled
Human (COCO)	10	13	0%	0	64.5%	54.7%	54.7%	4.2%
Model, short	11	14	0%	0	69.2%	62.1%	62.1%	5.3%
Model, detailed (cut at 77)	84	96.5	94%	25.2	80.7%	74.7%	71.5%	8.2%
Model, detailed, fitted	61	71.5	0%	0	80.7%	74.7%	74.7%	8.2%

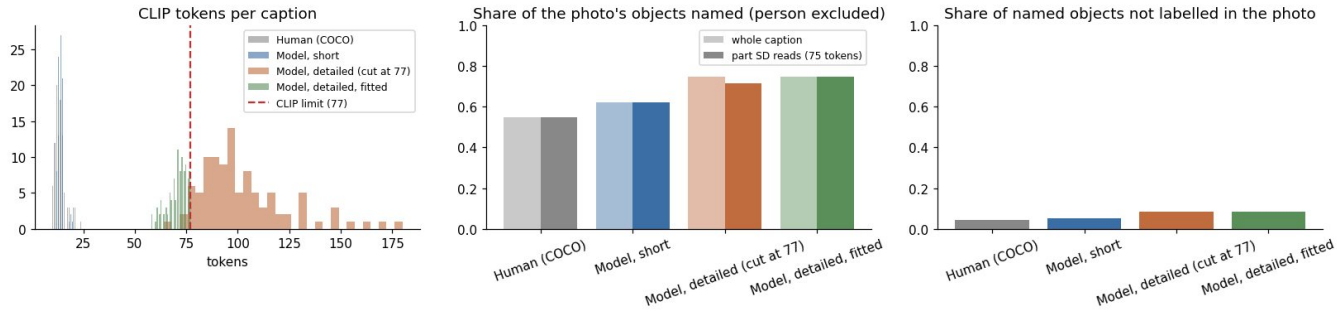


Figure 1. CLIP tokens per caption (red line: the 77-token limit); share of the photo's objects named, person excluded, in the whole caption (light) and in the part Stable Diffusion reads (dark); share of named objects not labelled in the photo.

Observations. With people left out, so the numbers line up with object recall, the gap between captions is wider: the human captions name 54.7% of the photo's other objects, the detailed ones 74.7%. Fitting removed a quarter of the words but none of the objects, and lets Stable Diffusion read all of them (74.7% vs 71.5% for the cut-off version). The cut-off cost only about 3 points of object information, because detailed captions mention the main objects early and spend their last sentences on scenery; this limits how large an effect the key test could show.

5. Are the images closer to the real scene?

Caption source	CLIP-I (to real photo)	CLIPScore, own prompt	CLIPScore, human caption	Object recall	Named-object recall	Person drawn when present
Human (COCO)	0.704	26.35	26.35	67.7%	88.5%	94.1%
Model, short	0.709	26.37	22.22	67.7%	88.0%	88.2%
Model, detailed (cut at 77)	0.716	27.03	21.86	62.6%	76.3%	90.2%
Model, detailed, fitted	0.718	27.18	22.31	66.0%	72.0%	92.2%
Real photos (detector ceiling)	—	—	—	91.5%	—	—

Difference (95% range)	Fitted vs cut-off detailed (key test)	Fitted vs human	Cut-off detailed vs human
CLIP-I	+0.002 (–0.008 to +0.011)	+0.014 (–0.001 to +0.029)	+0.012 (–0.002 to +0.027)
CLIPScore, own prompt	+0.16 (–0.33 to +0.66)	**+0.83 (+0.10 to +1.57)**	+0.67 (–0.09 to +1.40)
CLIPScore, human caption	+0.44 (–0.04 to +0.90)	**–4.05 (–4.87 to –3.24)**	**–4.49 (–5.35 to –3.61)**
Object recall (92–93 photos)	+0.033 (–0.043 to +0.113)	–0.018 (–0.097 to +0.056)	–0.051 (–0.142 to +0.039)
Named-object recall (62–81 photos)	–0.041 (–0.136 to +0.051)	–0.101 (–0.206 to +0.003)	**–0.124 (–0.239 to –0.013)**
Person drawn (51 photos)	+0.020 (–0.039 to +0.078)	–0.020 (–0.059 to 0.000)	–0.039 (–0.118 to +0.039)

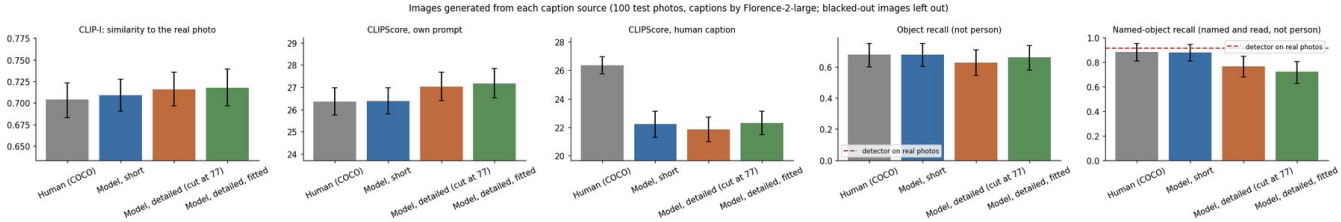


Figure 2. Image metrics by caption source with 95% ranges, blacked-out images left out; dashed lines mark the detector's recall on the real photos.

Observations. (1) **The key test shows no detectable difference** on any of the six measures: every 95% range includes zero. On object recall the range reaches +11 points, so a gain of that size cannot be ruled out with this

sample. (2) **No caption detectably beats the human one on the real scene**: CLIP-I creeps from 0.704 to 0.718 and object recall is never above the human caption's 67.7%. (3) The only detectable gain is **CLIPScore with the image's own prompt** for fitted captions (+0.83): images match the longer text slightly better, without getting closer to the photo. (4) Images from model captions score about 4 points lower against the **human** caption simply because they follow different text, so CLIPScore must always use the same reference caption when comparing models. (5) The detector finds 91.5% of objects in the real photos but 63–68% in generated images, whatever the caption.

6. Named-object recall: when a caption asks and the model reads

Object recall counts every object in the photo, including ones the caption never mentions. **Named-object recall** counts only the objects a caption names **in the part Stable Diffusion reads** and the photo shows (person excluded). A miss here cannot be blamed on the caption or on the token limit: the object was asked for and read.

Result. With human or short captions, 88% of the objects asked for and read are drawn. With detailed captions it is 76% (cut-off) and 72% (fitted). The drop against the human caption is **detectable for the cut-off detailed captions (–12.4 points, range –23.9 to –1.3)** and just short of detectable for the fitted ones (–10.1, range –20.6 to +0.3).

How to read it. Each object a long caption names is less likely to be drawn than an object a short caption names. Two explanations fit and this study cannot separate them: the model may spread its attention over more objects and draw each less reliably, or the extra objects that long captions name may simply be harder (smaller, secondary objects such as a cell phone or a fork). Either way, naming more objects does not translate into drawing more of them, which is consistent with object recall not rising.

7. "Named, read, but still missing" objects

Every object named in the part Stable Diffusion reads, shown in the real photo, and not found in the generated image was exported to **named_read_missing.csv** (image id, caption index, caption source, seed, caption, missing object, image path), for rescue experiments that rewind the generation to before the object is lost.

Caption source	Missing objects	…of which the detector finds in the real photo
Human (COCO)	9	9
Model, short	11	8
Model, detailed (cut at 77)	25	24
Model, detailed, fitted	34	33
Total	79 in 73 images	74

The most often missing objects are cell phone (9), bottle (8), bench (6), truck (6) and chair (5): mostly small or secondary objects. The 74 cases where the detector finds the object in the real photo are the most trustworthy, since the object's absence in the generated image is then unlikely to be a detector miss.

8. Examples



Figure 3. Four fixed example photos: the real photo, then the image from each of the four captions with the same seed (CLIP-I to the real photo in brackets; captions below each image).

Photo	What the examples show
Feet in orange pants on a bench	Every caption gives a whole person instead of the cropped legs. The two detailed captions give nearly identical images: removing the filler and the last sentences changed little.
Pig in a cage with a "Stop the pork" sign	Quoting the sign fills the image with garbled lettering. The fitted caption gives the clearest pig (CLIP-I 0.64 vs 0.57 cut-off), yet still on a street rather than in a cage.
Construction site with cones and a building	Both detailed captions get the building and truck together; neither draws the orange cones the caption names. The fitted version is slightly further from the photo (0.61 vs 0.64).
A batter swinging at a baseball game	The detailed captions add the catcher and the crowd, and both images show them; the human and short captions give a lone batter. All four images are black-and-white although the photo is in colour: no caption fixes that habit.

9. Conclusion

With this sample, longer captions gave no detectable gain. Captions that name 75% of a photo's objects instead of 55%, and fitted so Stable Diffusion reads all of them, produced images that were not detectably closer to the real photo and did not contain more of its objects. Fitting the detailed captions changed no measure detectably, and the cut-off turned out to remove mostly scenery rather than objects. Objects that long captions ask for are also drawn less reliably (named-object recall 72–76% vs 88%). Together these point to the model, not the captions or the token limit, as the main limit, although gains up to the top of the 95% ranges (about +11 points of object recall) remain possible until a larger run.

What this means for the project. Better or longer prompts are unlikely to close the gap between the 63–68% of objects Stable Diffusion draws and the 91.5% the detector finds in real photos; changing the model might, which supports the planned model-side experiments (LoRA fine-tuning and text-encoder connectors). Every comparison should judge images against the real photo and compute CLIPScore against the same reference captions. The 79 named-read-missing objects give the rescue experiments a ready set of clear failures.

10. Limitations

• 100 photos and one seed: the 95% ranges are wide; 300 photos and two seeds would narrow them. • Because the cut-off removed little object information (71.5% vs 74.7% without people), the key test could only detect a small effect. • Named-object recall is computed over different objects for each caption source, so its drop mixes model behaviour with harder objects. • The fitting rule ranks sentences by COCO objects named; it may drop useful non-object detail (actions, colours, positions). • Object recall relies on a detector that finds 91.5% of objects in real photos, and on COCO's 80 labelled object types. • Florence-2 was trained on large web datasets that may include COCO-like photos.

11. Reproducibility and files

The study ran entirely from scratch: the test split was rebuilt from the COCO annotations (matching the saved Milestone 1 split file), the 100 photos downloaded and cropped, captioned again, and all 400 images generated. The captions and caption statistics reproduced an earlier pilot exactly, and later runs that reused the saved captions and images reproduced every number.

File (results/recaptioning_scratch/)	Contents
captions.csv, captions_v2.csv	Human, short, detailed and fitted caption for each photo
caption_stats.csv, image_metrics.csv, differences.csv	The tables in Sections 4 and 5
generated_scores.csv	Every generated image with its scores, the objects named and read, the objects found, and a blacked-out flag
named_read_missing.csv	The named-read-missing list in Section 7
recaptioning_v2_summary.json	Every number in this report
runs/recaptioning_scratch/	The 400 generated images, one folder per caption source

Environment: Google Colab, Tesla T4, PyTorch 2.11, Diffusers 0.42.1, Transformers 5.22; models stable-diffusion-v1-5, openai/clip-vit-large-patch14, Florence-2-large, Faster R-CNN ResNet-50 FPN v2.